CBCS SCHEME

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SRI SAIRAM COLLEGE OF ENGINEERING

BCS501

Fifth Semester B.E./B.Tech. Degree Examination, June/July 2026
Software Engineering and Project Management

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	Explain software application domains.	10	L2	CO1
	b.	Explain software engineering layers.	05	L2	CO1
	c.	Explain software engineering myths.	05	L2	CO1
		OR			
Q.2	a.	Explain steps involved in generic software process.	10	L2	CO1
	b.	Explain the Waterfall Model and V-Model of software development process with a neat diagram.	10	L2	CO1
		Module – 2			
Q.3	a.	Explain steps involved in requirement engineering process.	10	L2	CO2
	b.	Explain tasks involved in establishing ground work for requirement engineering.	05	L2	CO2
	c.	Explain use case diagram for 'Safehome' home security function.	05	L2	CO2
		OR			
Q.4	a.	Explain the thumb rules that should be followed when creating analysis model.	06	L2	CO2
	b.	Explain elements of analysis model with a neat diagram.	06	L2	CO2
	c.	Explain the safe home security function with a neat sequence diagram.	08	L2	CO2
		Module – 3			
Q.5	a.	Explain the agile process.	10	L2	CO3
	b.	Explain extreme programming process with a neat diagram.	10	L2	CO3
		OR			
Q.6	a.	Explain the principles that guide practice.	10	L2	CO3
	b.	Explain the principles that guide communication.	10	L2	CO3
		Module – 4			
Q.7	a.	Explain the importance of project management.	06	L2	CO4
	b.	Explain the activities covered by software project management.	06	L2	CO4
	c.	Explain about Project Management Life Cycle.	08	L2	CO4
		OR			
Q.8	a.	Differentiate between Traditional versus Modern Project Management practices.	10	L2	CO4
	b.	Explain the cost benefit evaluation techniques.	10	L2	CO4
		Module – 5			
Q.9	a.	Explain about importance of software quality.	10	L2	CO5
	b.	Explain software quality and the software quality models.	10	L2	CO5
		OR			
Q.10	a.	Explain the observations on software project estimation.	10	L2	CO5
	b.	Explain the decomposition techniques.	10	L2	CO5

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BCS501

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2025/Jan.2026
Software Engineering and Project Management

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	Explain the domains of software applications.	08	L2	CO1
	b.	Outline the unique nature of WebApps.	08	L2	CO1
	c.	Explain various software myths. Discuss.	04	L2	CO1
		OR			
Q.2	a.	Explain the activities performed in a software process framework?	06	L2	CO1
	b.	Explain the waterfall model along with its pros and cons.	08	L2	CO1
	c.	Explain specialized process models.	06	L2	CO1
		Module – 2			
Q.3	a.	Explain how groundwork parameters are established in requirements engineering.	08	L2	CO2
	b.	What is the importance of quality function deployment in eliciting requirements?	06	L1	CO2
	c.	How can we validate requirements?	06	L1	CO2
		OR			
Q.4	a.	Explain about scenario based modelling.	10	L2	CO2
	b.	Illustrate regarding how can we create a Behavioral Model.	10	L2	CO2
		Module – 3			
Q.5	a.	Explain Agility along with the principles of Agility.	10	L2	CO3
	b.	Explain the Extreme Programming Process.	06	L2	CO3
	c.	Explain about the critics of XP.	04	L2	CO3
		OR			
Q.6	a.	Explain the scrum flow process.	08	L2	CO3
	b.	Explain the communication principles guiding framework activity.	08	L2	CO3
	c.	How can we validate and test principles in coding.	04	L1	CO3
		Module – 4			
Q.7	a.	Define Project. Show the contrast of software projects with other types of projects.	06	L2	CO4
	b.	Explain the ISO 12207 software development life cycle with a neat diagram.	10	L2	CO4
	c.	What are outsourced projects?	04	L1	CO4
		OR			
Q.8	a.	Illustrate the cost benefit evaluation techniques.	10	L2	CO4
	b.	Illustrate the concept of Risk evaluation.	10	L2	CO4
		Module – 5			
Q.9	a.	Explain the details to be drafted for achieving quality in software.	06	L2	CO5
	b.	Explain the software quality characteristics of ISO 9126.	08	L2	CO5
	c.	Explain process requirements for the process quality management.	06	L2	CO5
		OR			
Q.10	a.	Explain about the decomposition techniques.	10	L2	CO5
	b.	Explain the COCOMO II model.	10	L2	CO5

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CBCS SCHEME - Make-Up Exam
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BCS501

Fifth Semester B.E./B.Tech. Degree Examination, June/July 2025
Software Engineering and Project Management

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	Define Software Engineering. Briefly explain the seven broad categories of software application domains.	8	L2	CO1
	b.	Define software process. Explain five generic process framework activities.	6	L2	CO1
	c.	Explain any six principles that focus on software engineering practice as a whole.	6	L2	CO1
		OR			
Q.2	a.	Explain two common evolutionary process model with neat diagrams.	10	L2	CO1
	b.	Explain the different phases of the unified process model.	10	L2	CO1
		Module – 2			
Q.3	a.	Explain the seven distinct tasks of requirements engineering process.	10	L2	CO2
	b.	Develop a use case model for Safe Home system by considering home owner as a primary actor.	10	L3	CO2
		OR			
Q.4	a.	Develop an activity diagram for eliciting requirements.	6	L3	CO2
	b.	Explain the different requirement modeling approaches.	10	L2	CO2
	c.	How does analysis clauses manifest themselves as elements of the solution spaces. Mention its different ways.	4	L2	CO2
		Module – 3			
Q.5	a.	Explain the by activities of Xp process with a neat diagram.	10	L2	CO3
	b.	Explain the following agile process models. i) Adaptive Software Development (ASD) ii) Scrum.	10	L2	CO3
		OR			
Q.6	a.	Explain any five principles i) That guide process ii) That guide practice.	10	L2	CO3
	b.	Explain the ten different planning principles.	10	L2	CO3
		Module – 4			
Q.7	a.	Explain the sequence of software development activities recommended by ISO12207 with a neat diagram.	10	L2	CO4
	b.	Make a list of different activities that management involves. Also explain the principal project management processes along with a list of different activities that management involves.	10	L2	CO4

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		OR			
Q.8	**a.**	Explain the traditional versus modern project management practices.	**10**	**L2**	**CO4**
	b.	Explain the different aspects of evaluation of individual projects.	**10**	**L2**	**CO4**
		Module – 5			
Q.9	**a.**	Explain the following software quality models. i) Garvin's quality dimensions ii) McCall's Model.	**10**	**L2**	**CO5**
	b.	Explain the place of software quality step wise frame work in project planning.	**10**	**L2**	**CO5**
		OR			
Q.10	**a.**	Write a short note on : i) Software siting Approaches ii) Problem based estimation.	**10**	**L2**	**CO5**
	b.	Write a short note on : i) Dromey's Model ii) Boehm's Model.	**10**	**L2**	**CO5**

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BCS501

Fifth Semester B.E./B.Tech. Degree Examination, June/July 2025
Software Engineering & Project Management

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	Explain the software process in software engineering highlighting the importance of software engineering.	10	L2	CO1
	b.	Explain the five activities that a generic process framework for software engineering encompasses.	10	L2	CO1
		OR			
Q.2	a.	Explain software myths with examples.	10	L2	CO1
	b.	Explain Incremental process models and evolutionary process models with a neat diagram.	10	L2	CO1
		Module – 2			
Q.3	a.	Explain the different tasks which requirements engineering encompasses.	10	L2	CO2
	b.	Explain the nature and characteristics of software system.	10	L2	CO2
		OR			
Q.4	a.	Explain requirements elicitation and various techniques used in requirements elicitation along with its importance.	10	L2	CO2
	b.	Illustrate an UML use case diagram for home security function.	10	L2	CO2
		Module – 3			
Q.5	a.	Explain Agile process and agility principles.	10	L2	CO3
	b.	Explain Extreme Programming (XP) with a neat diagram.	10	L2	CO3
		OR			
Q.6	a.	Explain SCRUM process with a neat diagram.	10	L2	CO3
	b.	Explain Agility with the cost of change with diagram. Explain the principles of Agile software development.	10	L2	CO3
		Module – 4			
Q.7	a.	Explain different categories of software projects with example.	10	L2	CO4
	b.	Compare between Project Management Life Cycle And Software Development Life Cycle and its phases.	10	L2	CO4
		OR			
Q.8	a.	Explain the difference between traditional and modern project management.	10	L2	CO4
	b.	Explain the concepts in activity planning in software project management.	10	L2	CO4
		Module – 5			
Q.9	a.	Explain place of software quality in project management.	10	L2	CO5
	b.	Explain in detail the techniques to enhance software quality.	10	L1	CO5
		OR			
Q.10	a.	Explain Quality Management Systems. With principles of BSENISO9001 : 2000.	10	L2	CO5
	b.	Explain the techniques to enhance software quality and software reliability. Explain SEICMM levels.	10	L2	CO5

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BCS501

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Software Engineering and Project Management

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	Explain software process and software engineering practices.	10	L2	CO1
	b.	Explain the waterfall model and incremental model, with diagram.	10	L2	CO1
		OR			
Q.2	a.	Explain Boehm Spiral process model with a neat diagram. Mention its advantages and disadvantages.	10	L2	CO1
	b.	Explain the five activities of a generic process framework for software engineering.	10	L2	CO1
		Module – 2			
Q.3	a.	Explain the distinct tasks of requirement engineering.	10	L2	CO2
	b.	Illustrate the UML use case diagram for safe home system.	10	L2	CO2
		OR			
Q.4	a.	Explain Class-Responsibility-Collaborator(CRC) modeling and data modeling with an example.	10	L2	CO2
	b.	Explain the elements of analysis model in requirement modeling.	10	L2	CO2
		Module – 3			
Q.5	a.	Explain the principles of agile process development.	10	L2	CO3
	b.	Explain the following : i) Adaptive software development ii) SCRUM	10	L2	CO3
		OR			
Q.6	a.	Explain the concepts of extremes programming with a neat diagram.	10	L2	CO3
	b.	Explain design modeling principles that guide the respective framework activity.	10	L2	CO3
		Module – 4			
Q.7	a.	Illustrate the project management life cycle with a neat diagram.	10	L2	CO4
	b.	Explain : i) Different ways of categorizing software projects ii) Smart objectives	10	L2	CO4
		OR			
Q.8	a.	Explain the difference between traditional versus modern project management practices along with the role of management.	10	L3	CO4
	b.	Explain software development life cycle (ISO 12207) with a neat diagram.	10	L2	CO4
		Module – 5			
Q.9	a.	Explain Quality Management System with principles of BS EN ISO-9001-2000.	10	L2	CO5
	b.	Explain the following : i) McCall model ii) Garvin's Quality Dimensions.	10	L2	CO5
		OR			
Q.10	a.	Describe six generic functions allowed in automated estimation techniques of software projects.	10	L3	CO5
	b.	Explain COCOMO II model.	10	L2	CO5

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